Year 9

Mini Test 4

Geometry

Baldivis Secondary College

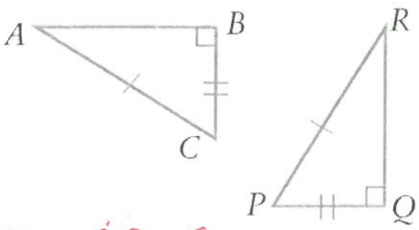
Total Marks: / 16
Total Time: 30 minutes

Name: ______________________

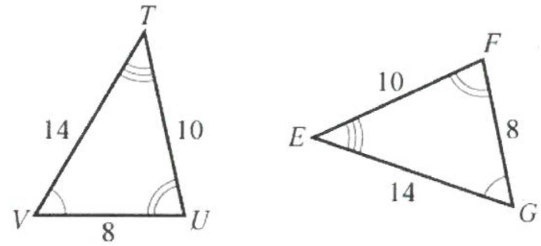
Notes Pages and Calculators allowed

Question 1 — 15 ~~6~~ **marks**

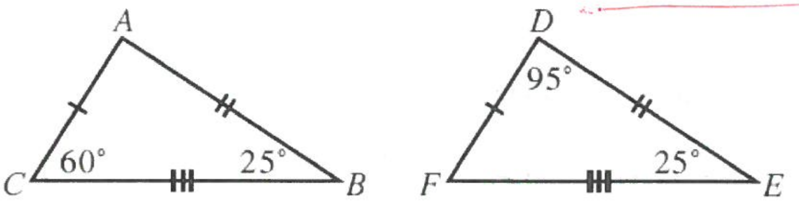
Prove that the following triangles are congruent:
Remember to write down the rule e.g: AAA, SSS, SAS



$PQ = CB$ ✓
$AC = PR$ ✓
$\angle B = \angle Q$ ✓ (RHS rule) ✓
$\Delta ABC \cong \Delta RQP$ ✓



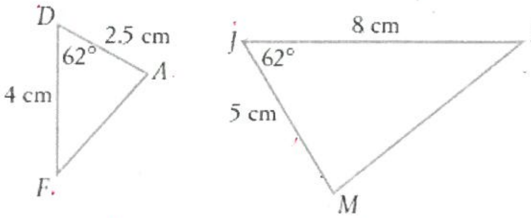
$\angle T = \angle E$ ✓ $VU = GF$
$\angle V = \angle G$ ✓ $FE = UT$ (5 marks)
$\angle U = \angle F$ ✓ $EG = TV$
$\Delta TVU \cong \Delta EGF$ (SSS or AAA rule)

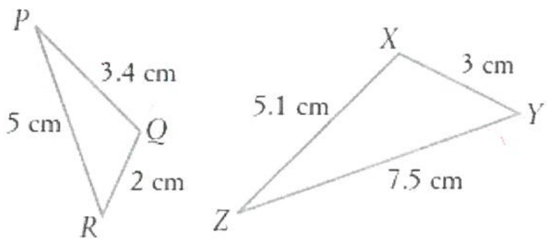


$\angle B = \angle E$ — $60° + 25° = 85$
$ED = BA$ — $180 - 85 = 95°$
$FE = CB$ — $180 - 120 = 60°$ (5 marks)
$DF = AC$ — $\angle C = \angle F$
$\angle A = \angle D$
(SSS or AAA) $\Delta ACB \cong \Delta DFE$

Question 2 — 10 ~~4~~ **marks**

Determine if each of the following pairs of triangles are similar:
Remember to write down the rule e.g: AAA, SSS, SAS





$\angle FDA = \angle MJK$ ✓

$\frac{DA}{JM} = \frac{2.5}{5} = \frac{25}{50} = \frac{1}{2}$ ✓

$\frac{DF}{JK} = \frac{4}{8} = \frac{1}{2}$ ✓

✓ need to get order right
$\Delta FDA \sim \Delta MJK$
(SAS rule) ✓

$\Delta PQR \sim \Delta ZXY$ ✓
(SSS rule) ✓

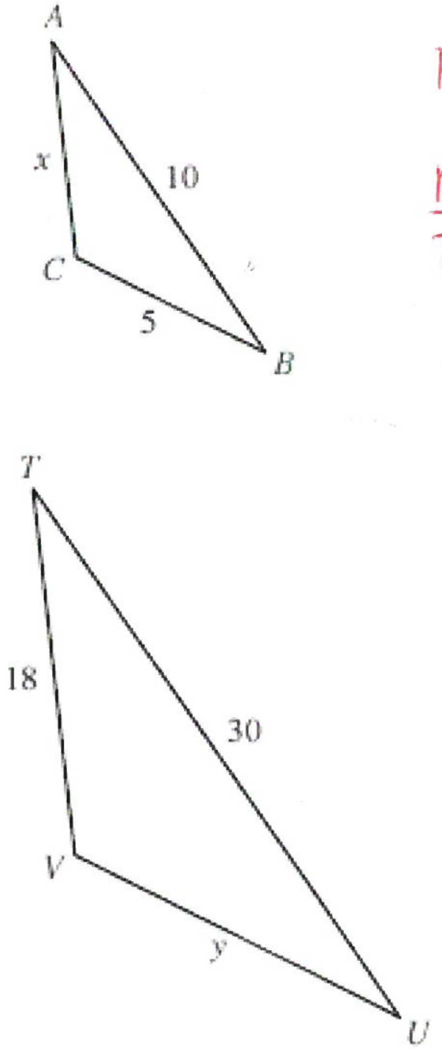
✓ $\frac{QR}{XY} = \frac{2}{3}$

✓ $\frac{PR}{ZY} = \frac{5}{7.5} = \frac{50}{75} = \frac{2}{3}$

✓ $\frac{PQ}{ZX} = \frac{3.4}{5.1} = \frac{34}{51} = \frac{2}{3}$

Question 3	6 marks

A. Find the unknown side for the following similar triangles:



Find x

$\frac{AC}{TV} = \frac{AB}{TU}$ ✓

$\frac{x}{18} = \frac{10}{30}$ ✓

$x = \frac{10}{30} \times 18 = 6$ ✓

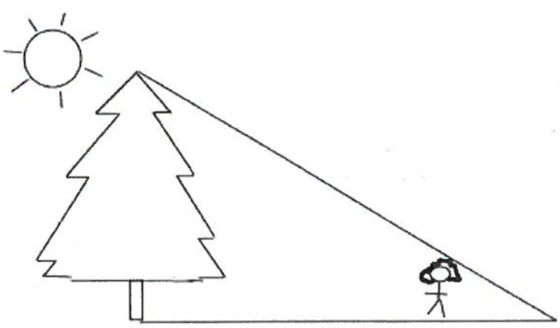
Find y

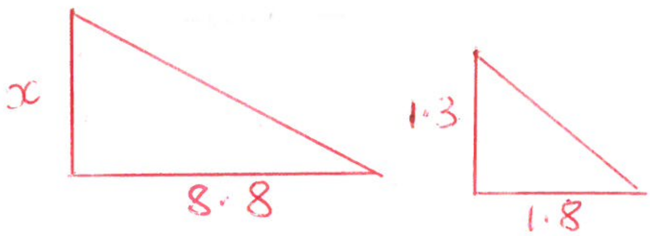
$\frac{UV}{BC} = \frac{UT}{BA}$ ✓

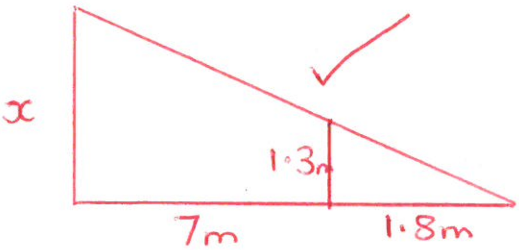
$\frac{y}{5} = \frac{30}{10}$ ✓

$y = \frac{30}{10} \times 5 = 15$ ✓

B. Tonya is 1.3 meters tall. She stands 7 meters in front of a tree and casts a shadow 1.8 meters long. How tall is the tree?







$\frac{x}{8 \cdot 8} = \frac{1 \cdot 3}{1 \cdot 8}$ ✓

$x = \frac{1 \cdot 3}{1 \cdot 8} \times 8 \cdot 8$

$x = 6 \cdot 35m$ ✓